

though the concept of digitisation helped during the pandemic, human intervention could not be avoided.

Here is where RefillBot, a Bengaluru based startup, steps in with its IoT solution. Started in the year 2020, RefillBot is a B2B company that provides a set of IoT hardware and software which can fit into the infrastructure of utilities. The data is automated across a unified dashboard and can be used to simplify, improve, and scale operations.

The company has three product lines: water monitoring, payments, and sustainable packaging. The utility that is currently being focused on is water. “Even though there are implementational, operational, and maintenance agencies, water management companies require a person to handle the data on a day-to-day basis. Our solution helps avoid even this cost for these agencies.”

Many water plants that have been set up by the Indian state and Central governments in more than 10,000 villages operate on manual basis. RefillBot’s pilot project has made it easy for these governments to get the data without any manual intervention. The company caters to both private and public enterprises.

Within just eight months, RefillBot has been deployed in hundred villages, with no negative reports. They are currently ‘live’ in 100 villages across seven *talukas* in Karnataka, including 35 to 40 *gram panchayats*. The private enterprises are implementing the system in another 100 villages. Beverage manufacturers have also shown interest in RefillBot’s solution.

RefillBot puts data at an abstract level that can be directly used for production. The hardware of the system includes flow meters, power sensors, and TDS monitors. This hardware is production-ready, so that the data comes to the cloud on which innovators can build their own software. An API layer on top is also provided. The pilot-level production so far includes



Founders of RefillBot (L to R): Savitri, Naveen, and Prabhu

the most essential parameters only, but these will increase with the new versions being developed.

The entire hardware for the system was arranged indigenously for the first prototype in use, which is running since eight months without any trouble. When it comes to communication, the company uses telemetry between the data path servers. The data is compressed and communicated over SMS.

RefillBot’s USP is its internet-free operation, which helps because the signal strength is very low in Indian villages. The GSM based pattern is hashed as the key is present only with the end device and servers. Encryption and decryption is done with security algorithms like SHA256 and MD5. Even in case of decryption in between, the data would not benefit anyone.

As many companies are going through a digital transformation for the first time, there is no upfront cost or investment needed. The minimum duration for a contract is six months on a pay-as-you-go basis. The contract includes the entire hardware, software, maintenance, and warranties/guarantees.

“Our device may look generic, something that can be easily built by a normal communication network, but our patent is for the core base

and the pluggable model. We do not have any specific certifications as we believe in rather spending money on customer needs. We are pretty sure that our devices will pass the certifications needed,” says Naveen.

The company invites innovators, service companies, entrepreneurs, etc to improve the platform. “City based local entrepreneurs/innovators can build solutions and deploy them locally. Our consumer platform hardware is open to use and open to solutions being built on it,” says Savitri.

The challenges overcome by RefillBot are many, as it started off during the pandemic. Starting with the hardware a plus-one-model of components had to be used for an industry standard device. The next was refining the circuit in terms of telemetry and security algorithms for data tracking. RefillBot is thankful to Arrow Electronics partnership and QuickTells help who sourced components in time and refined the circuit in terms of telemetry.

When it comes to practical challenges, each of the 100 remote villages presented their own challenge. “Amidst lack of consistent power supply and constant weather changes, our devices had to be tested. Yet, within a span of 45 days, our projects were live,” says Prabhu. **EFY**

—Sharon Abhignya Katta